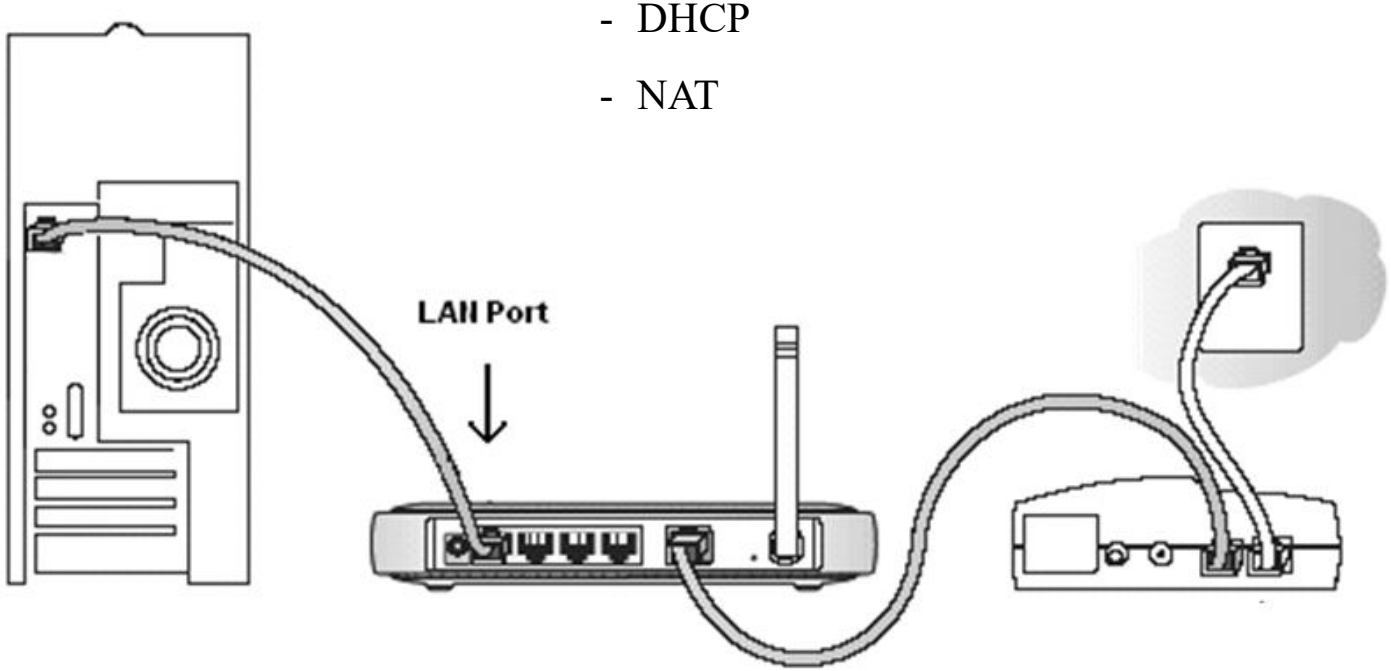
NAT(Network Address Translation)

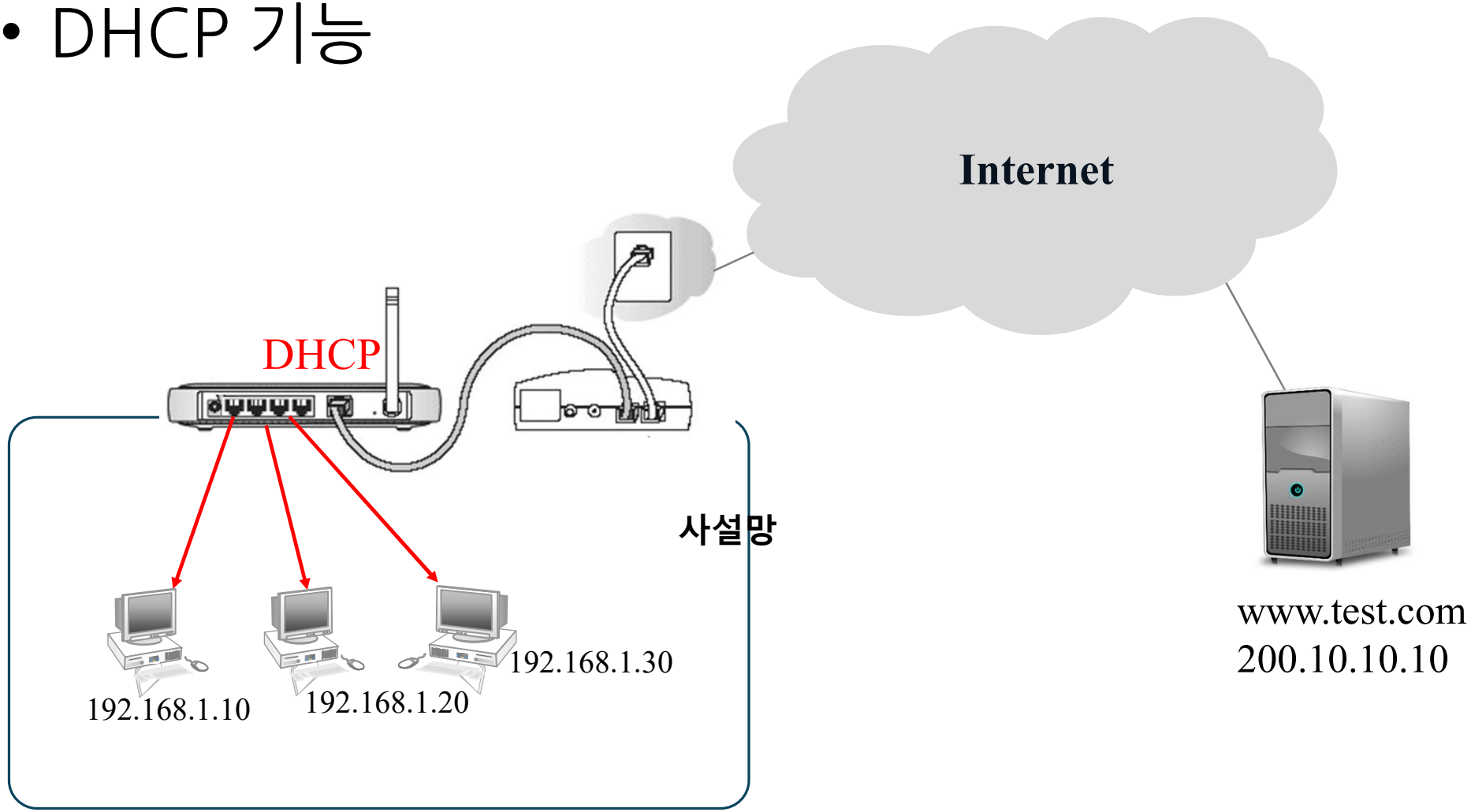
SoHo(Small office/home office) Network

- 공유기 기능
  - Routing
  - DHCP
  - NAT



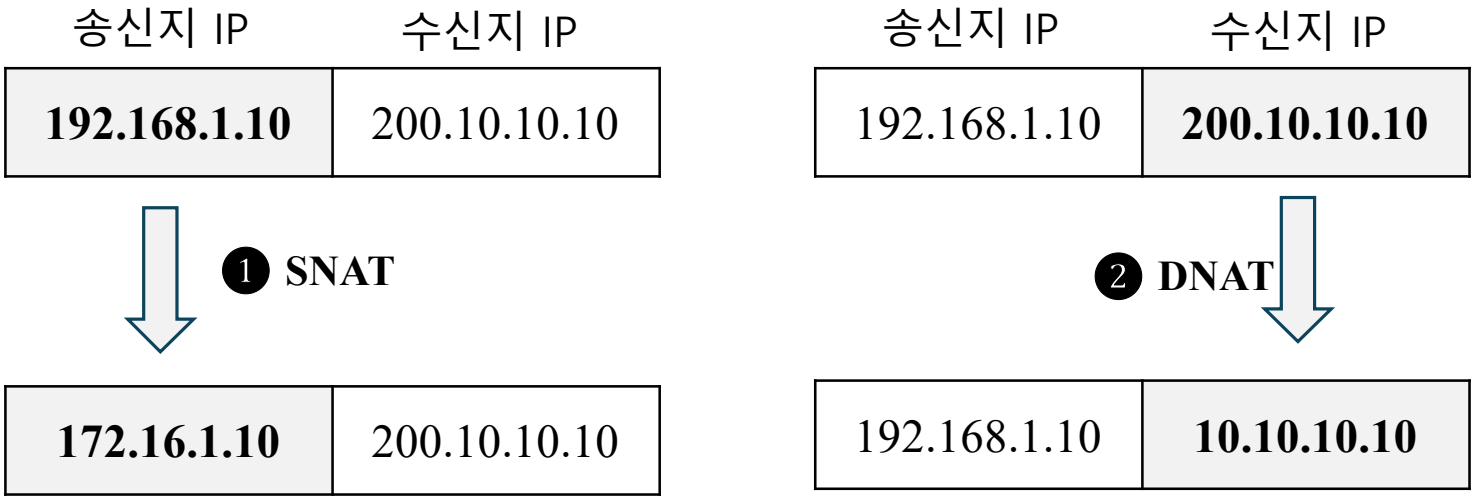
❶ 컴퓨터 ❷ 유/무선 공유기 ❸ 모뎀

- DHCP 기능


Internet
DHCP
사설망
192.168.1.10
192.168.1.20
192.168.1.30
www.test.com
200.10.10.10

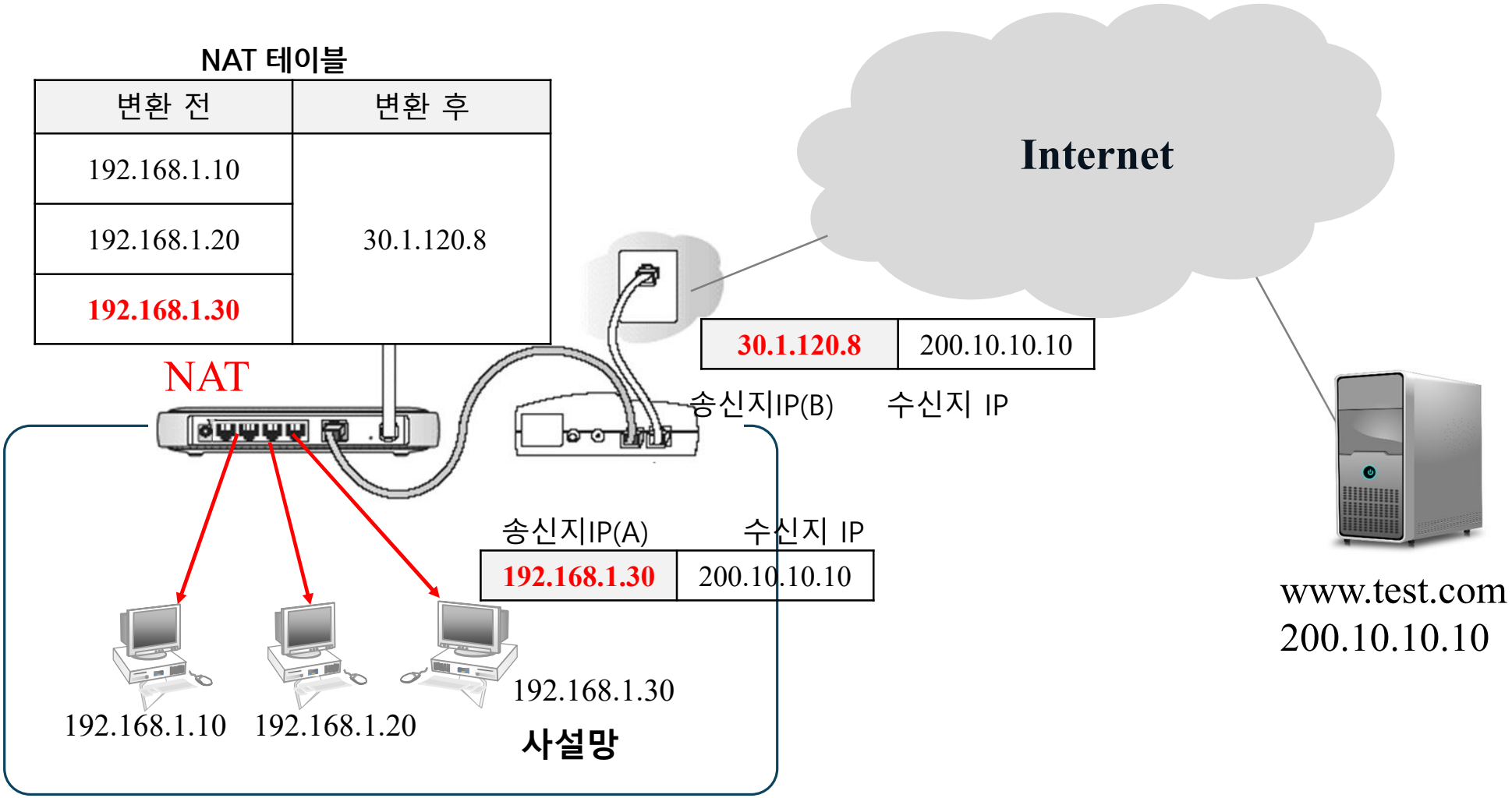
• NAT 기능

- 네트워크 주소 변환 (Network Address Translation)
- 송신지 또는 수신지 IP address를 다른 주소로 변환
- Source NAT(SNAT)와 Destination NAT(DNAT)



NAT 테이블

변환 전	변환 후
192.168.1.10	30.1.120.8
192.168.1.20	
192.168.1.30	

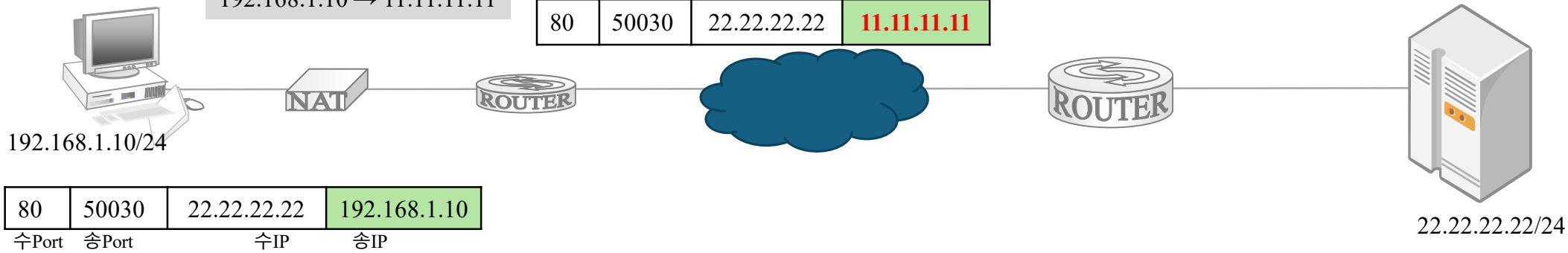


Source NAT

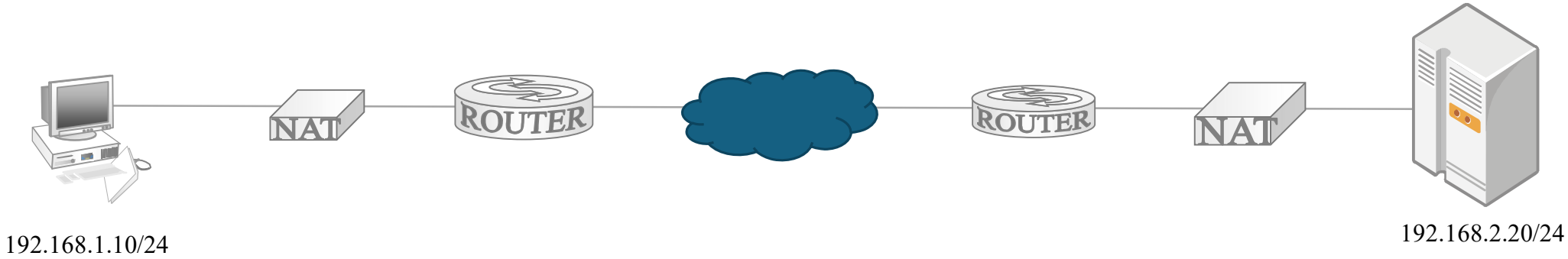
NAT Table

192.168.1.10 → 11.11.11.11

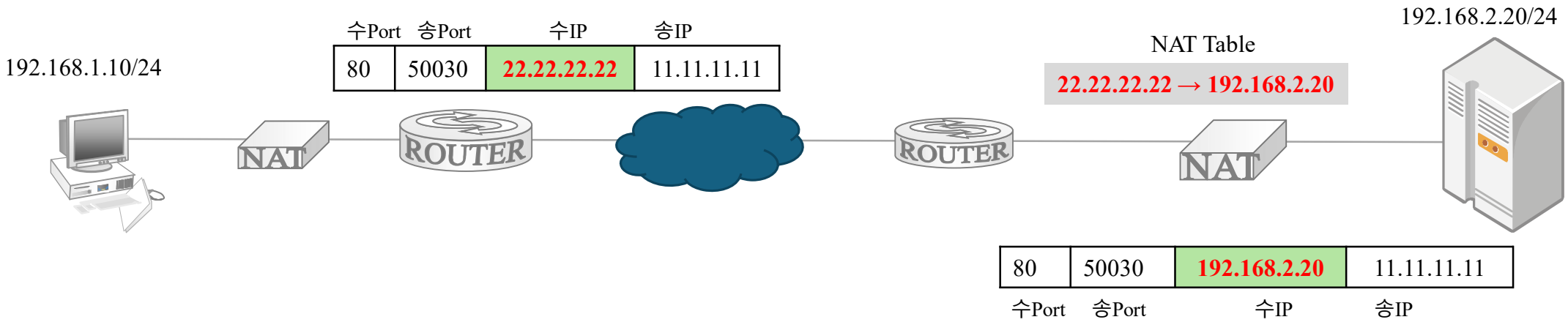
수Port	송Port	수IP	송IP
80	50030	22.22.22.22	11.11.11.11



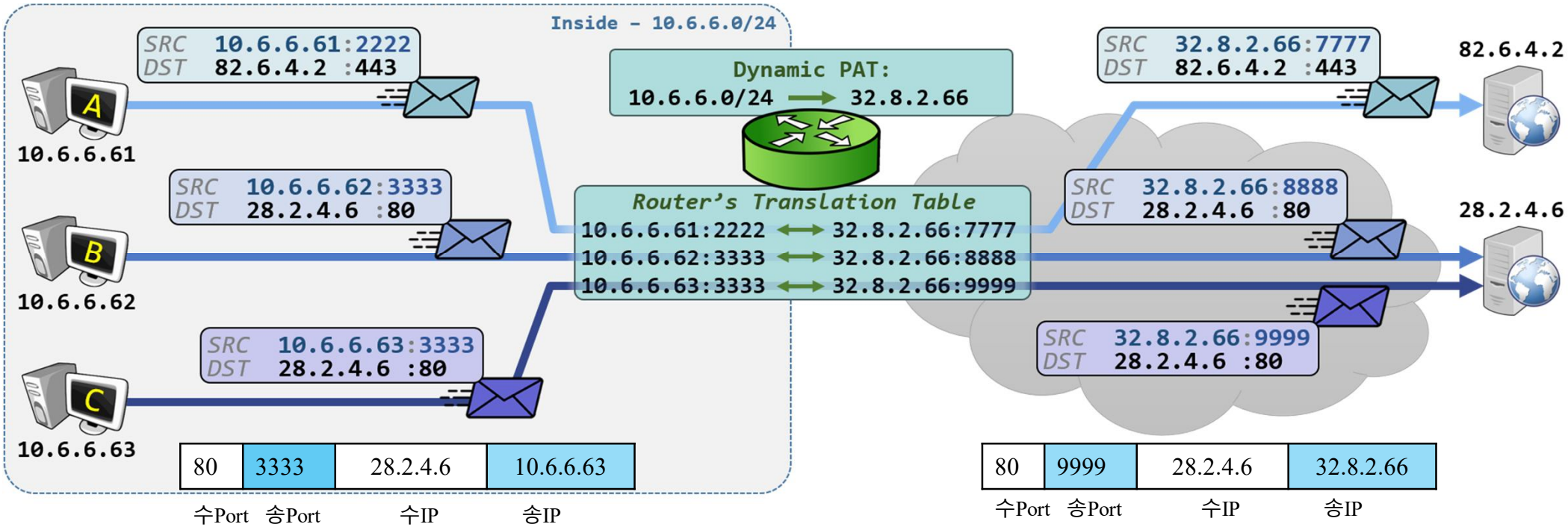
80	50030	22.22.22.22	192.168.1.10
수Port	송Port	수IP	송IP

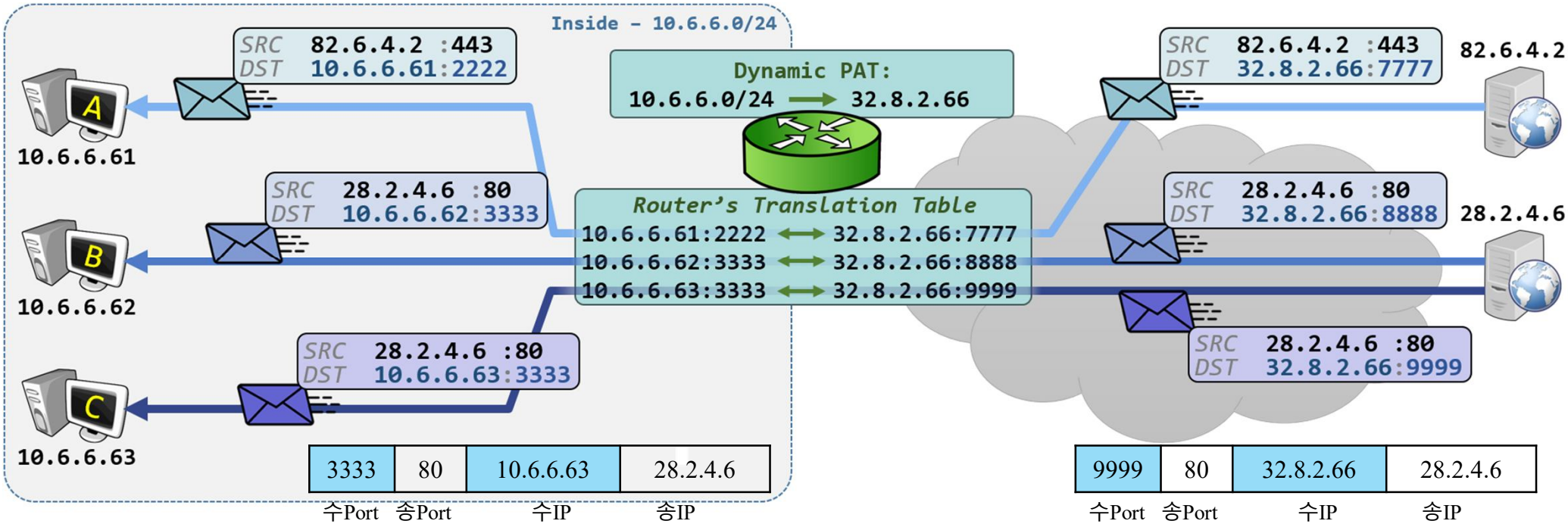


Destination NAT



PAT(Port Address Translation)




Inside - 10.6.6.0/24
SRC 82.6.4.2 :443
DST 10.6.6.61:2222
A
10.6.6.61
SRC 28.2.4.6 :80
DST 10.6.6.62:3333
B
10.6.6.62
SRC 28.2.4.6 :80
DST 10.6.6.63:3333
C
10.6.6.63
Dynamic PAT:
10.6.6.0/24 → 32.8.2.66
Router's Translation Table
10.6.6.61:2222 ↔ 32.8.2.66:7777
10.6.6.62:3333 ↔ 32.8.2.66:8888
10.6.6.63:3333 ↔ 32.8.2.66:9999
SRC 82.6.4.2 :443
DST 32.8.2.66:7777
82.6.4.2
SRC 28.2.4.6 :80
DST 32.8.2.66:8888
28.2.4.6
SRC 28.2.4.6 :80
DST 32.8.2.66:9999
3333 | 80 | 10.6.6.63 | 28.2.4.6
수Port 송Port 수IP 송IP
9999 | 80 | 32.8.2.66 | 28.2.4.6
수Port 송Port 수IP 송IP

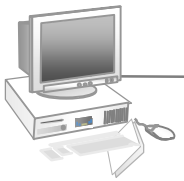
Port Mapping (Port Forwarding)

수Port	송Port	수IP	송IP
80	50010	200.1.1.10	10.10.10.10

수Port	송Port	수IP	송IP
5000	50010	192.168.1.10	10.10.10.10

HTTP
192.168.1.10:5000

http://200:1.1.10



10.10.10.10/24

FTP
192.168.1.20:5001

RDP
192.168.1.30:5002

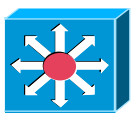
<<Port Fording Table>>

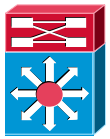
외부주소	내부주소
200.1.1.10:**80**	**192.168.1.10:5000**
200.1.1.10:21	192.168.1.20:5001
200.1.1.10:3389	192.168.1.30:5002

Multilayer Switch

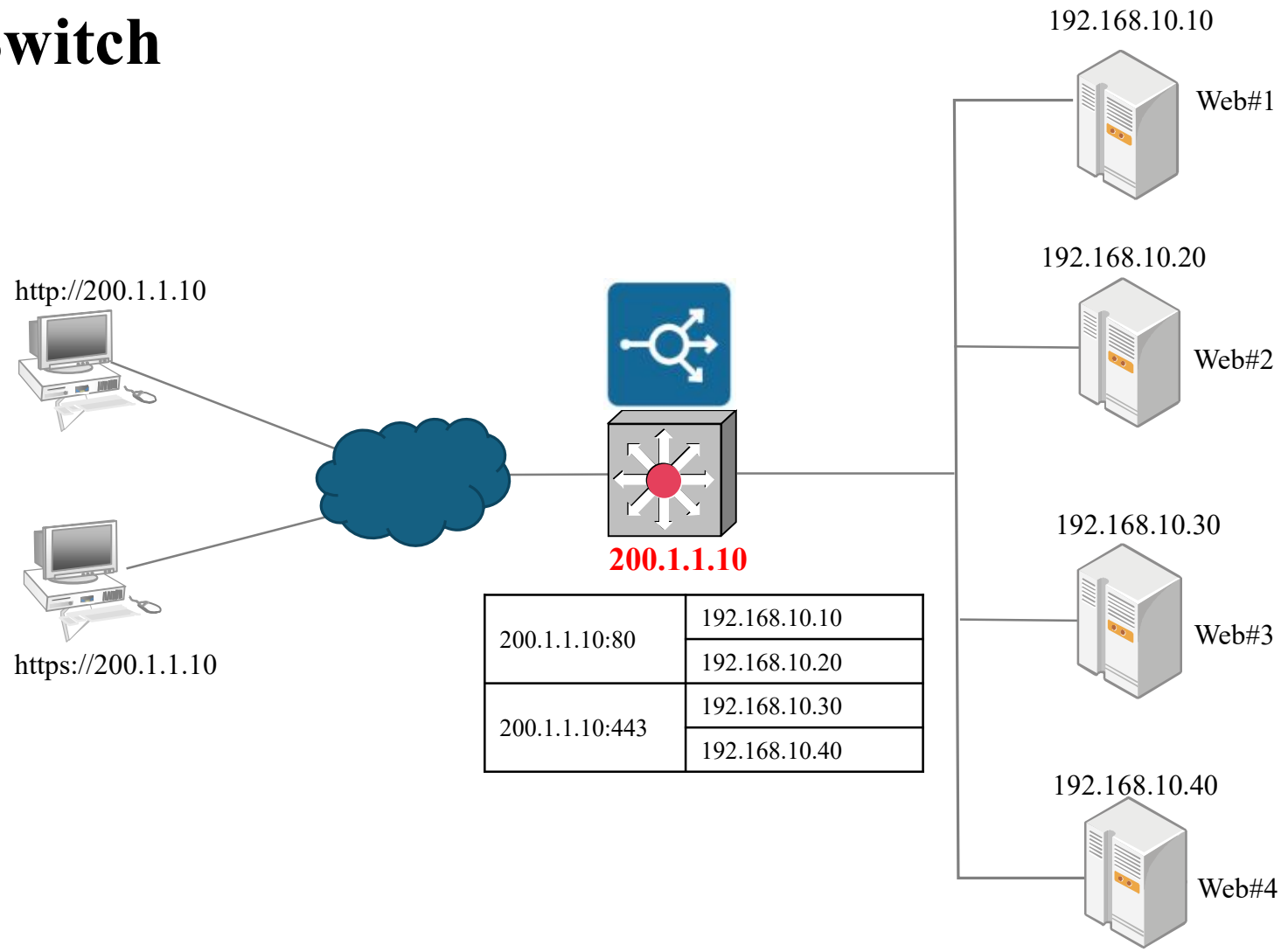
Multilayer Switch(다계층스위치)

구분	L2 Switch	L3 Switch	L4 Switch	L7 Switch
포워딩기준	2계층 (MAC Address)	3계층 (IP Address)	4계층 (IP address + Port 번호)	7계층 (IP address + Port 번호+ Content)
기능	Switching - Learning - Forwarding - Filtering	Switching Routing	L3 Switch Load Balance	L4 Switch Security Content 인식
주요 용도	Frame 전송	Packet 전송	FLB SLB	FLB SLB Security



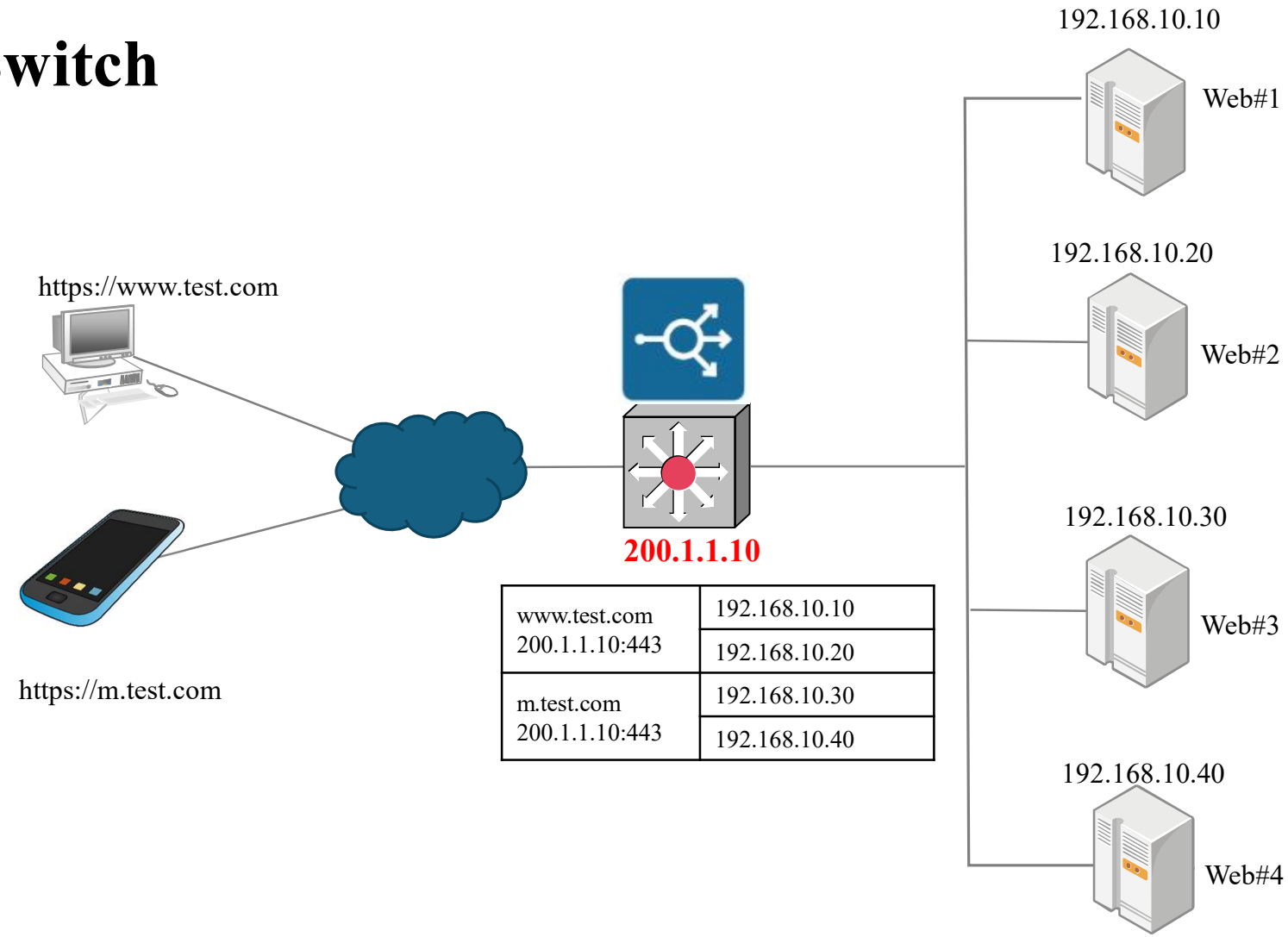


L4 Switch



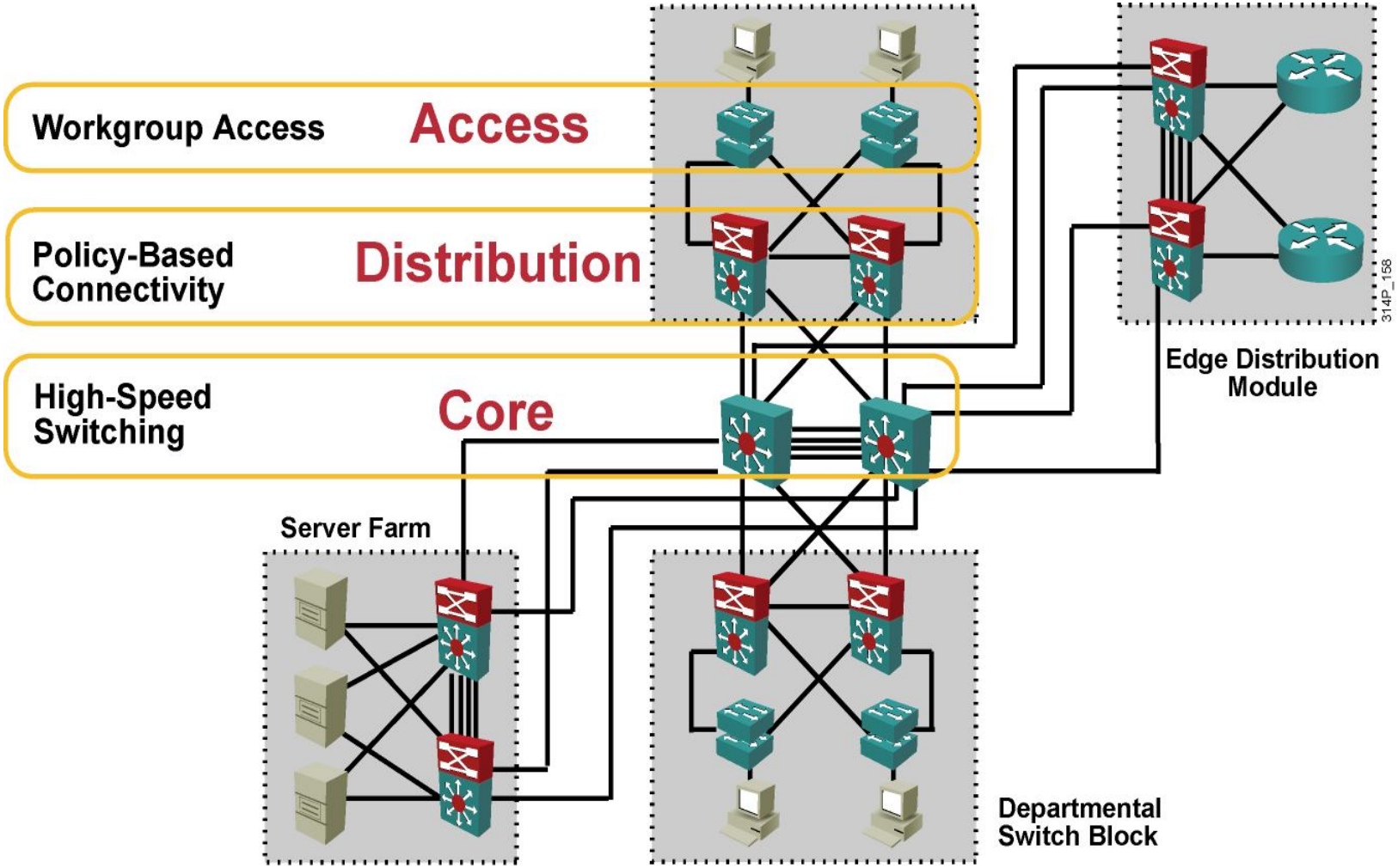
200.1.1.10:80	192.168.10.10
	192.168.10.20
200.1.1.10:443	192.168.10.30
	192.168.10.40

L7 Switch

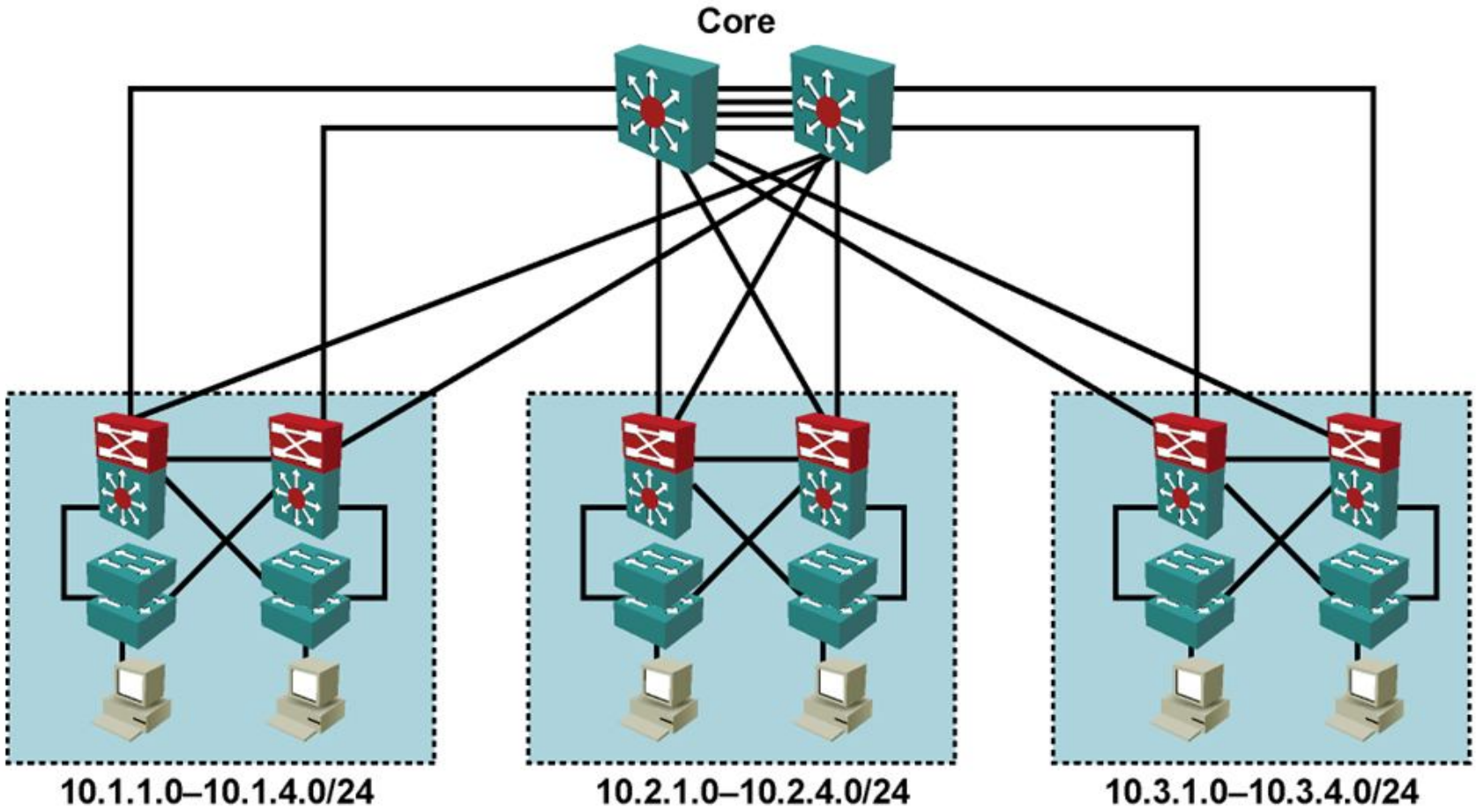


www.test.com 200.1.1.10:443	192.168.10.10
	192.168.10.20
m.test.com 200.1.1.10:443	192.168.10.30
	192.168.10.40

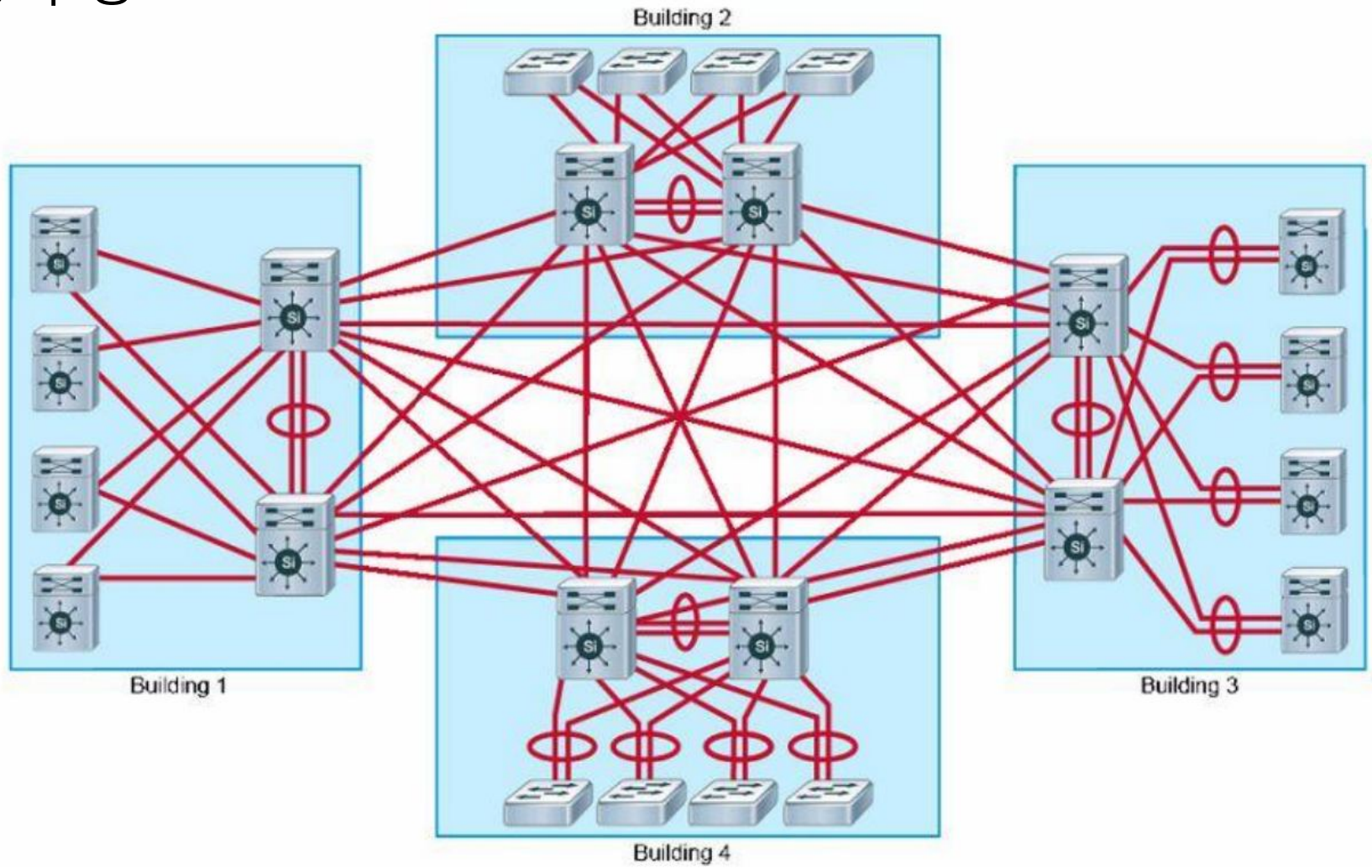
❶ 구성도


Workgroup Access
Access
Policy-Based Connectivity
Distribution
High-Speed Switching
Core
Server Farm
Edge Distribution Module
Departmental Switch Block
314P_158

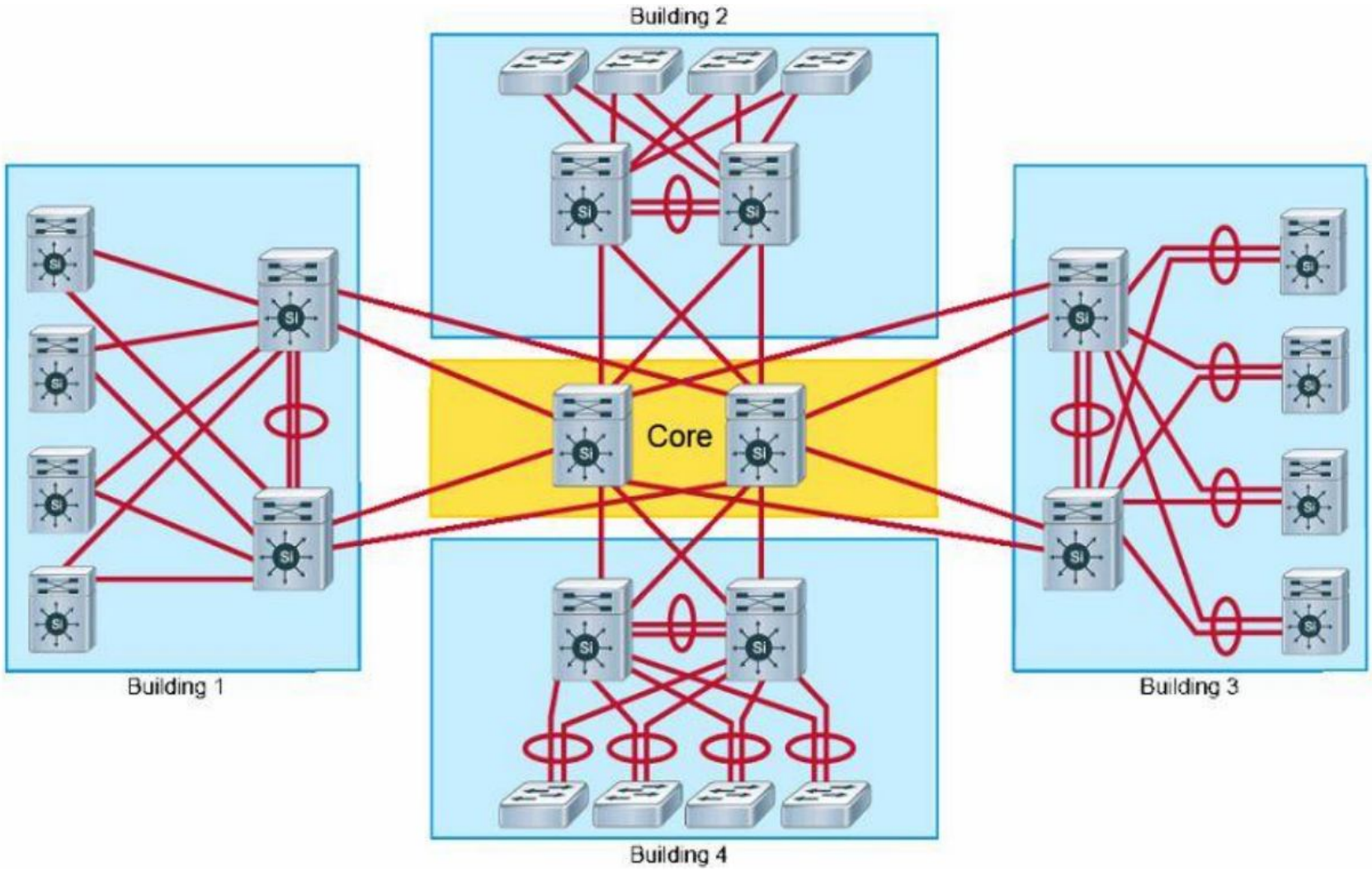
❷ 구성도



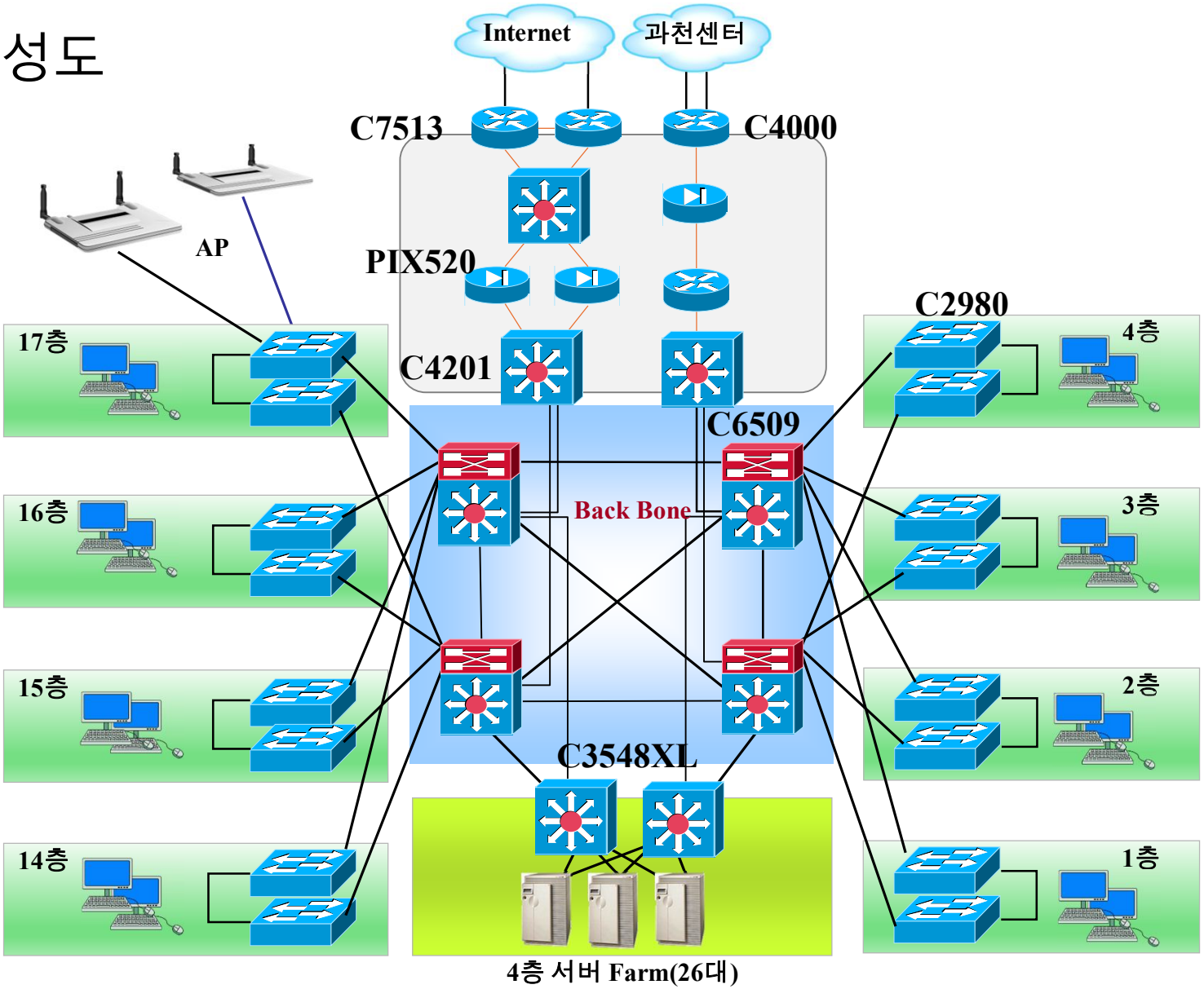
❸ 구성도



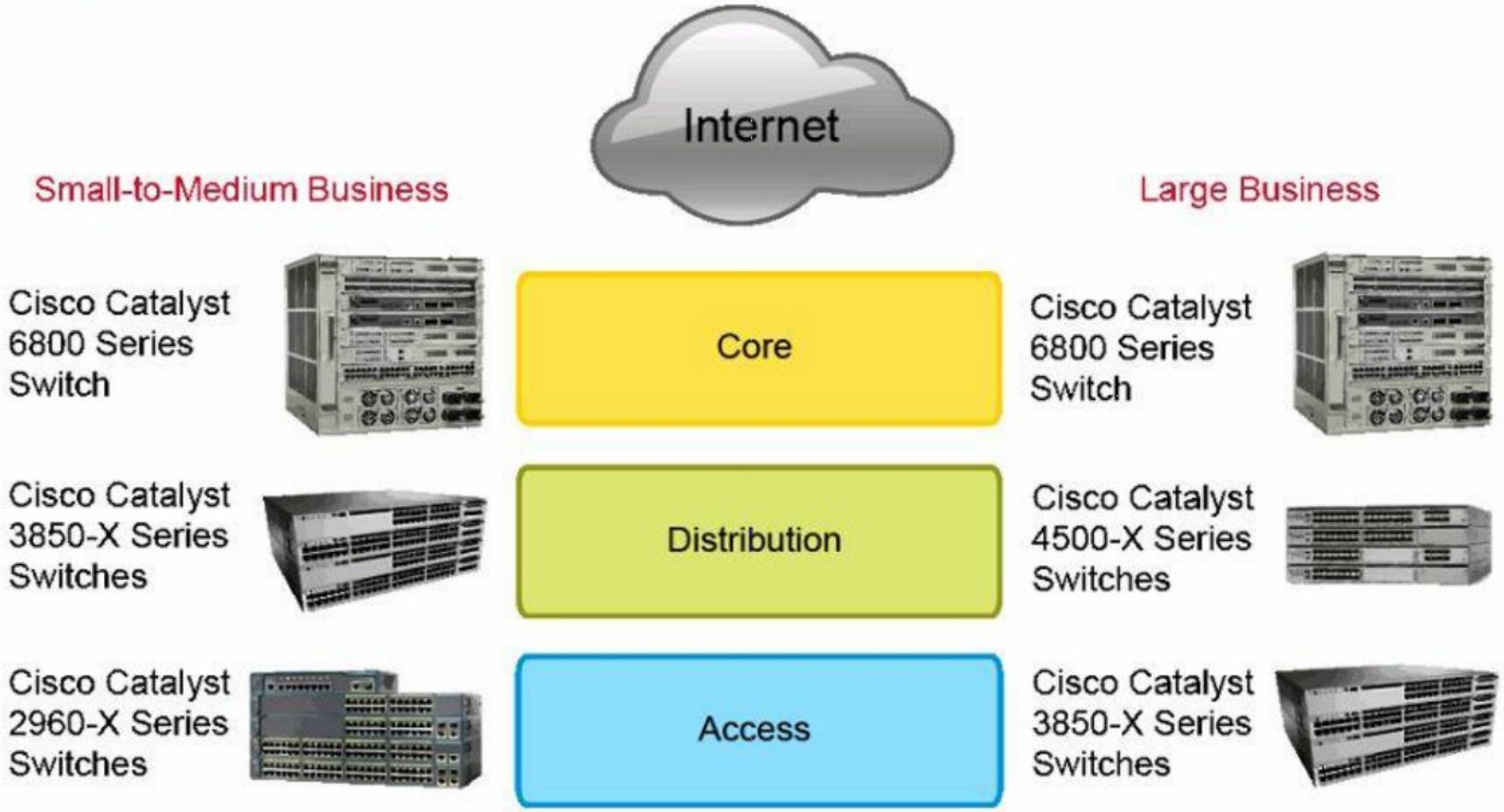
❹ 구성도



❺ 구성도

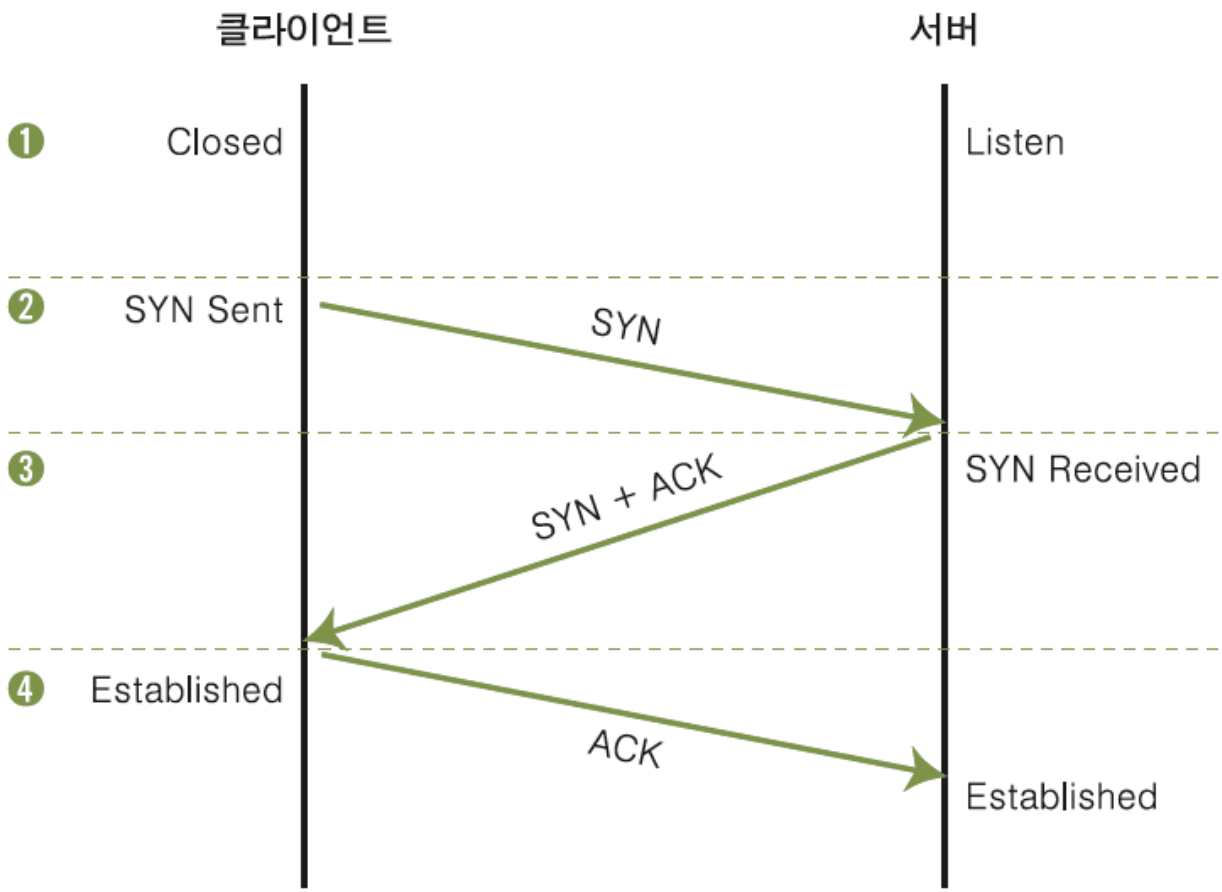

Internet
과천센터
C7513
C4000
AP
PIX520
C4201
C6509
C2980
Back Bone
C3548XL
17층
16층
15층
14층
4층
3층
2층
1층
4층 서버 Farm(26대)

- Cisco Multilayer Switches

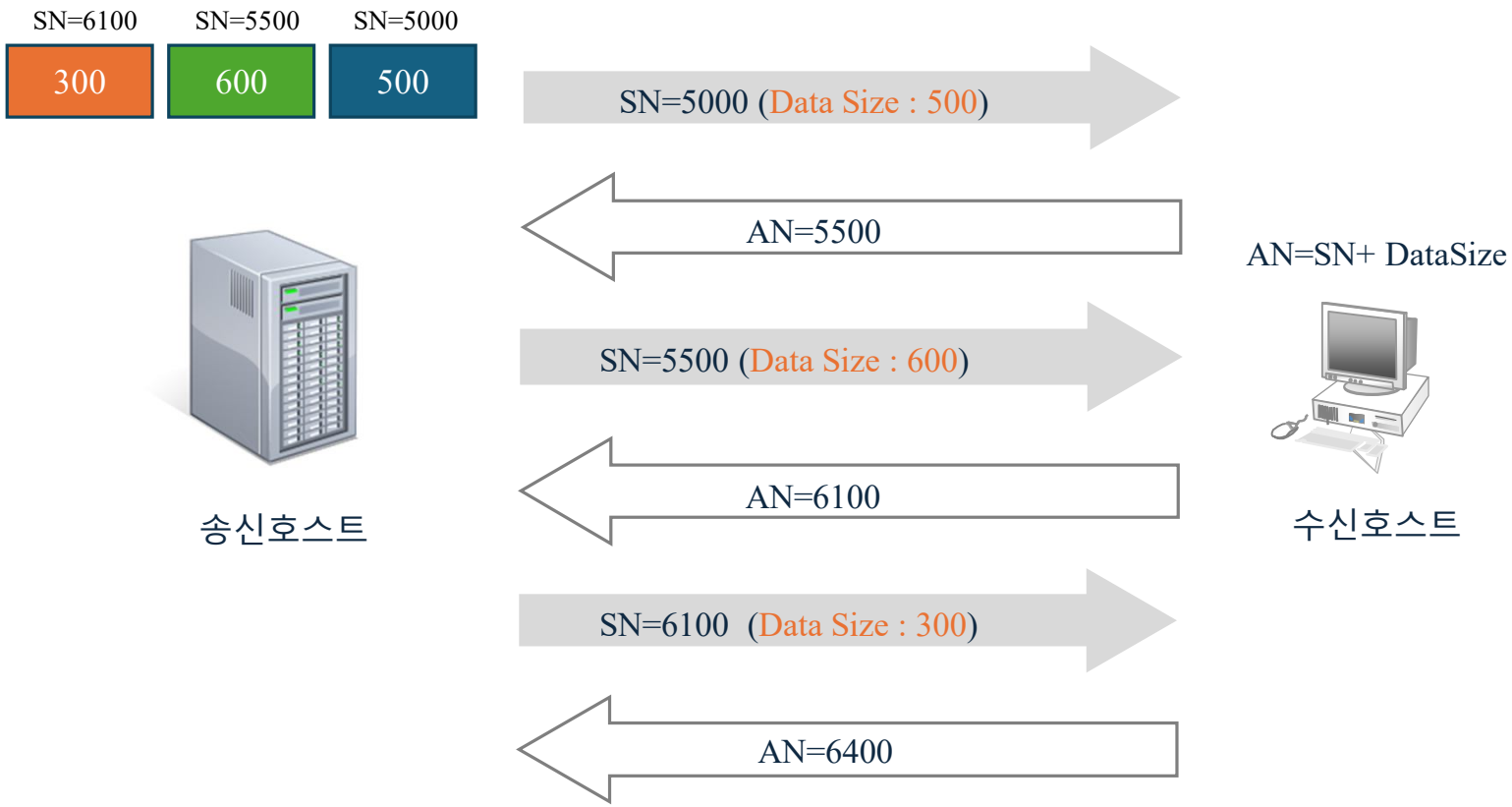


TCP

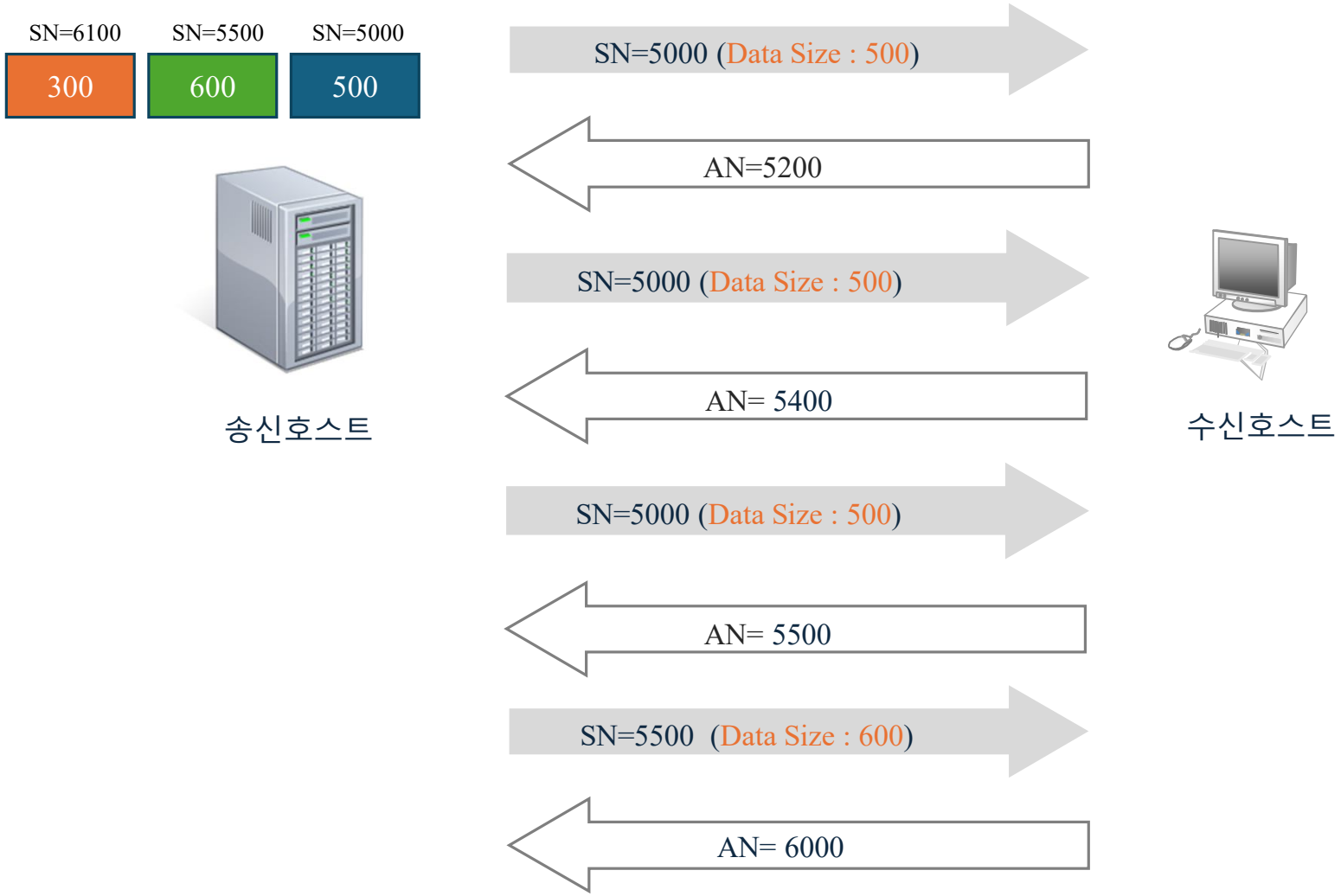
TCP 연결 설정 (3-Way Handshaking)



❶ 정상적인 트래픽 전송 과정

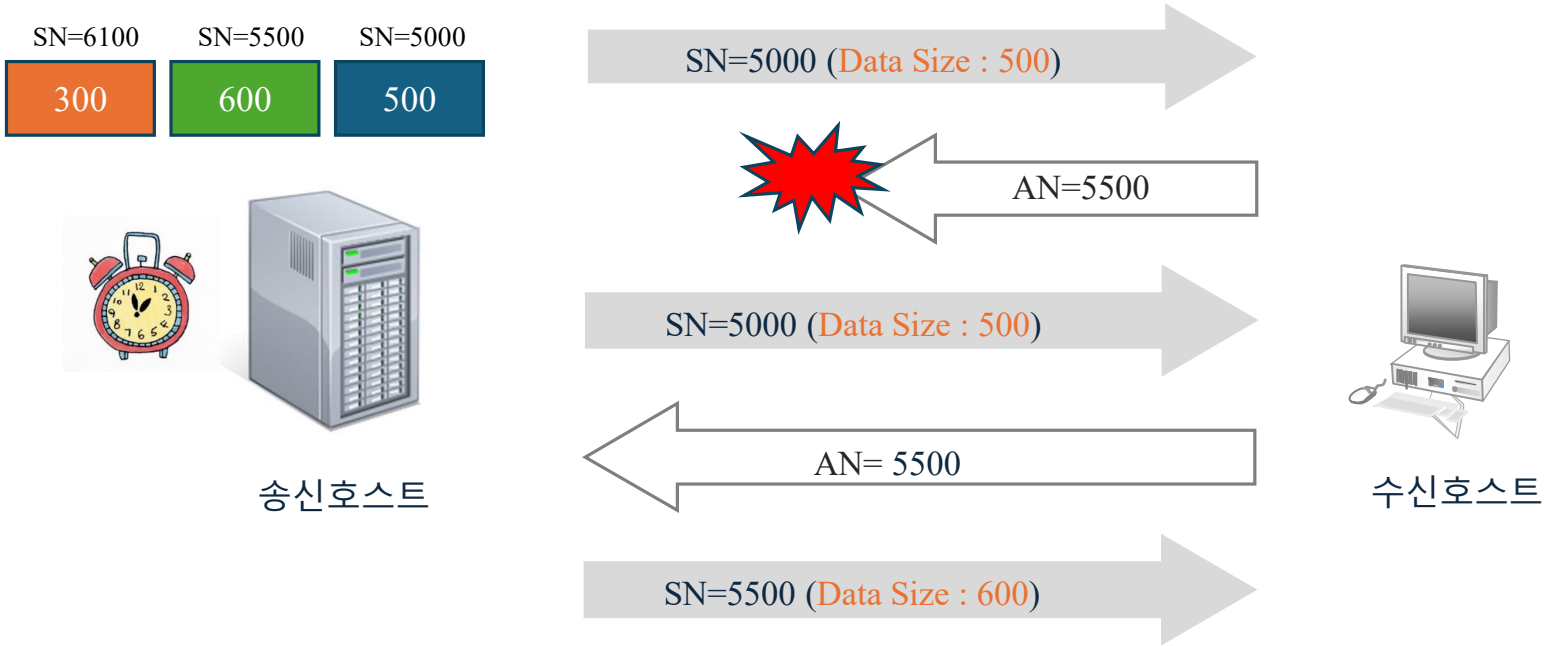


❷ 비정상적인 트래픽 전송 과정

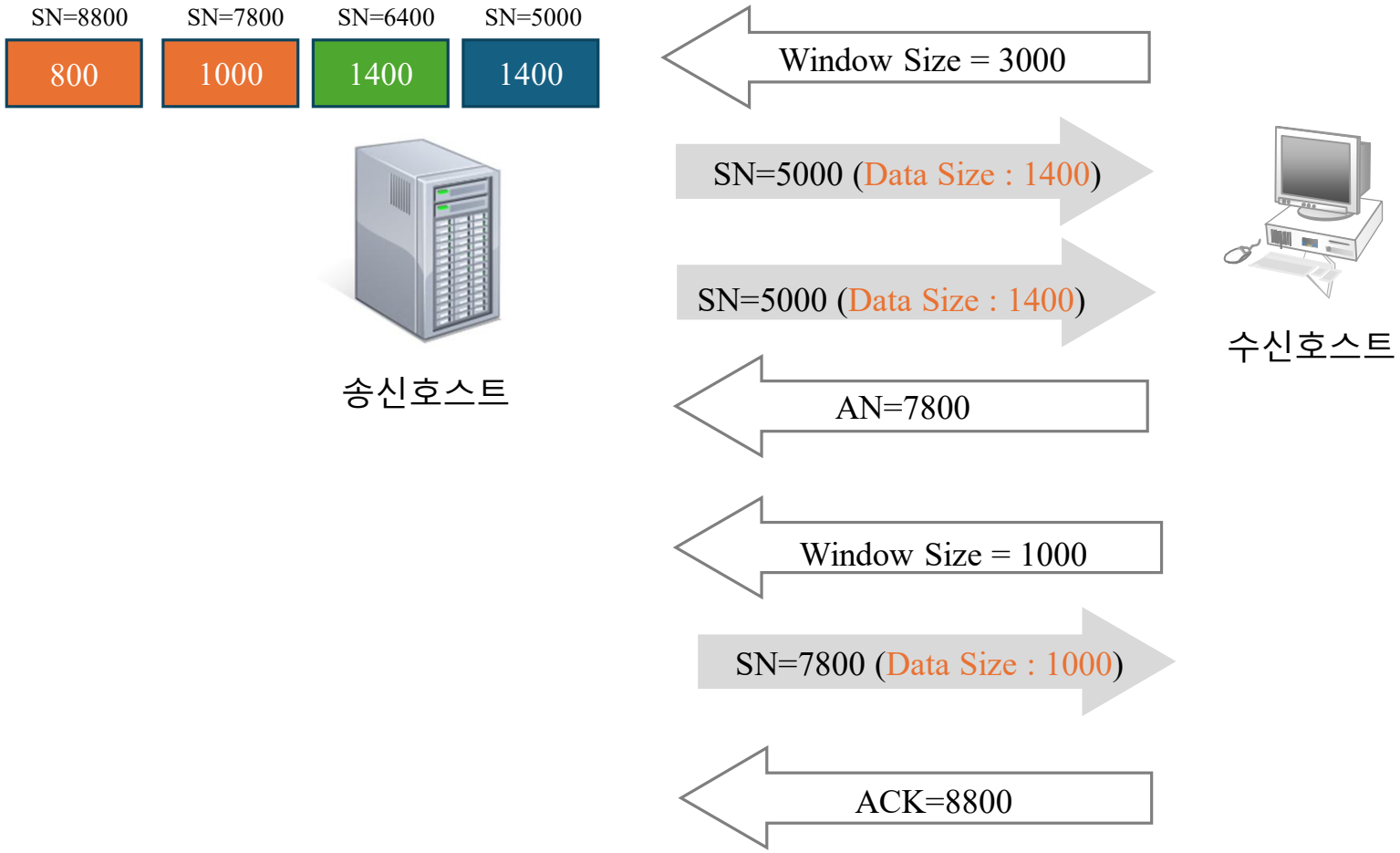


❷ 비정상적인 트래픽 전송 과정

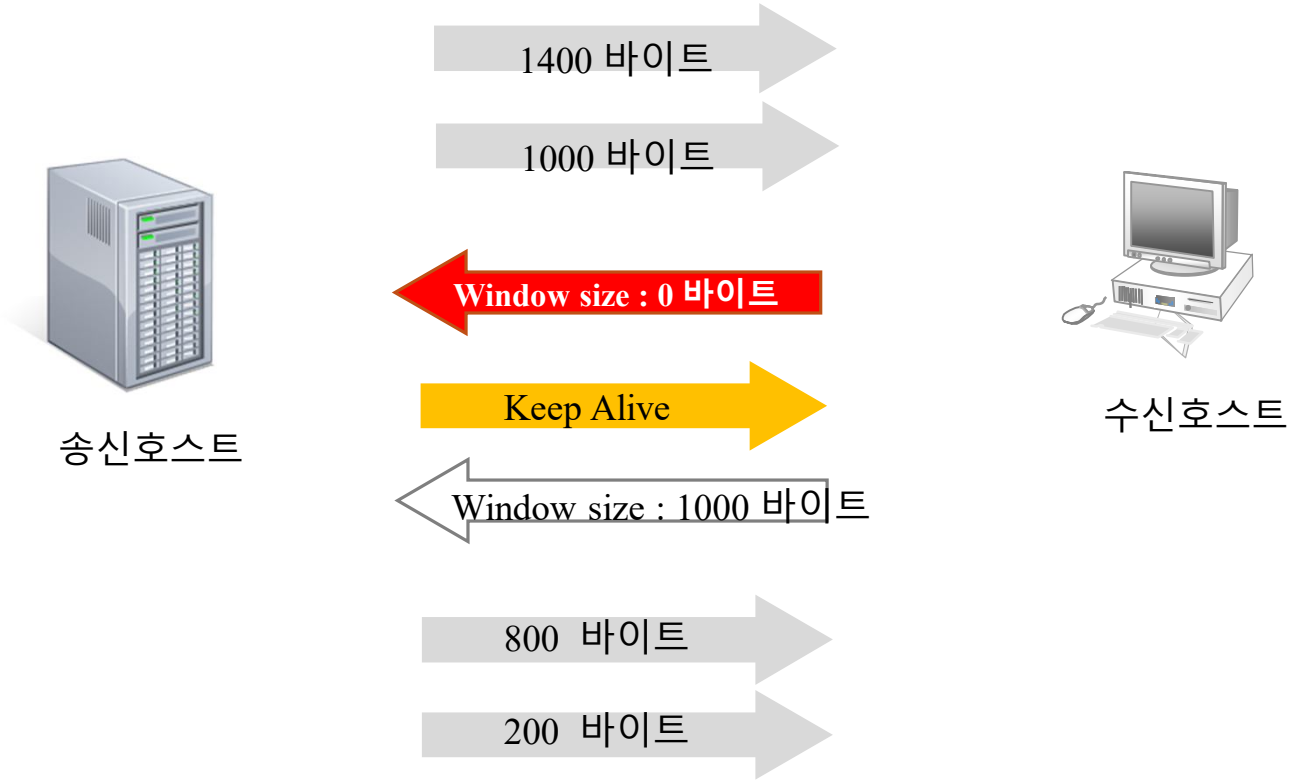
(예) 송신측에서 재전송 시도



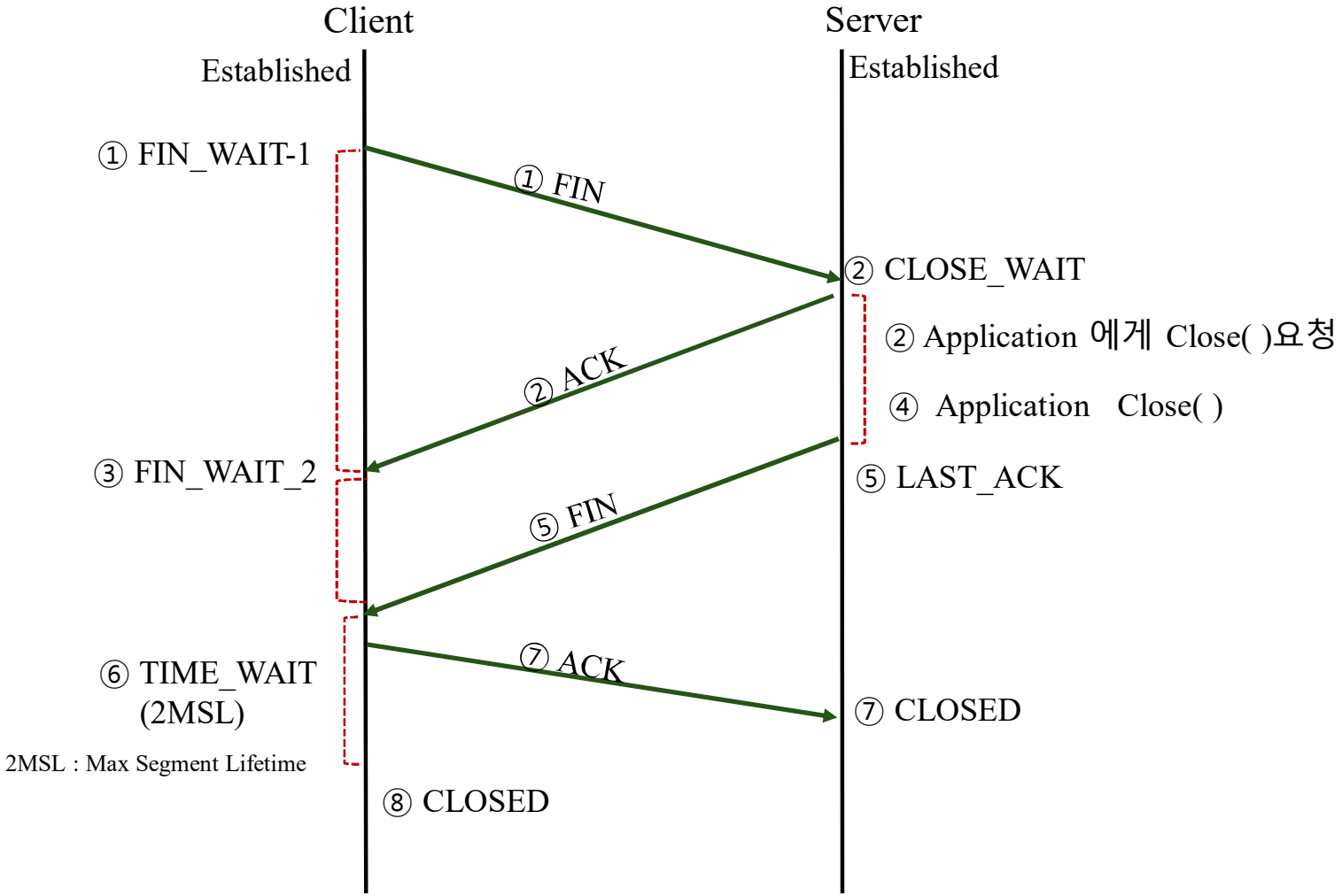
트래픽 흐름제어


SN=8800
SN=7800
SN=6400
SN=5000
800
1000
1400
1400
Window Size = 3000
SN=5000 (Data Size : 1400)
SN=5000 (Data Size : 1400)
AN=7800
Window Size = 1000
SN=7800 (Data Size : 1000)
ACK=8800
송신호스트
수신호스트

트래픽 흐름제어


1400 바이트
1000 바이트
Window size : 0 바이트
Keep Alive
Window size : 1000 바이트
800 바이트
200 바이트
송신호스트
수신호스트

TCP 연결 종료(4-way Handshake)

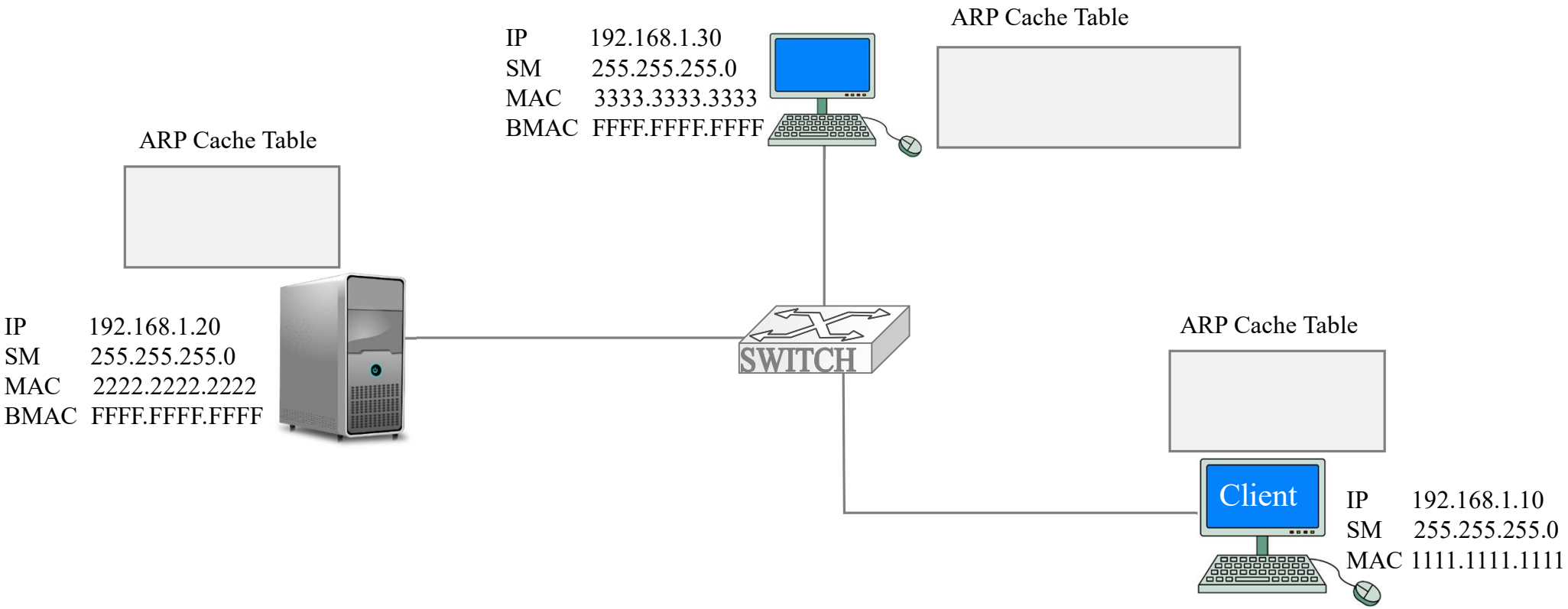


ARP cache table update

ARP 캐시 테이블(Cache Table)

- IP주소와 MAC 주소의 대응 관계를 저장한 테이블
- ARP 캐쉬 테이블 확인 명령어 : arp -a

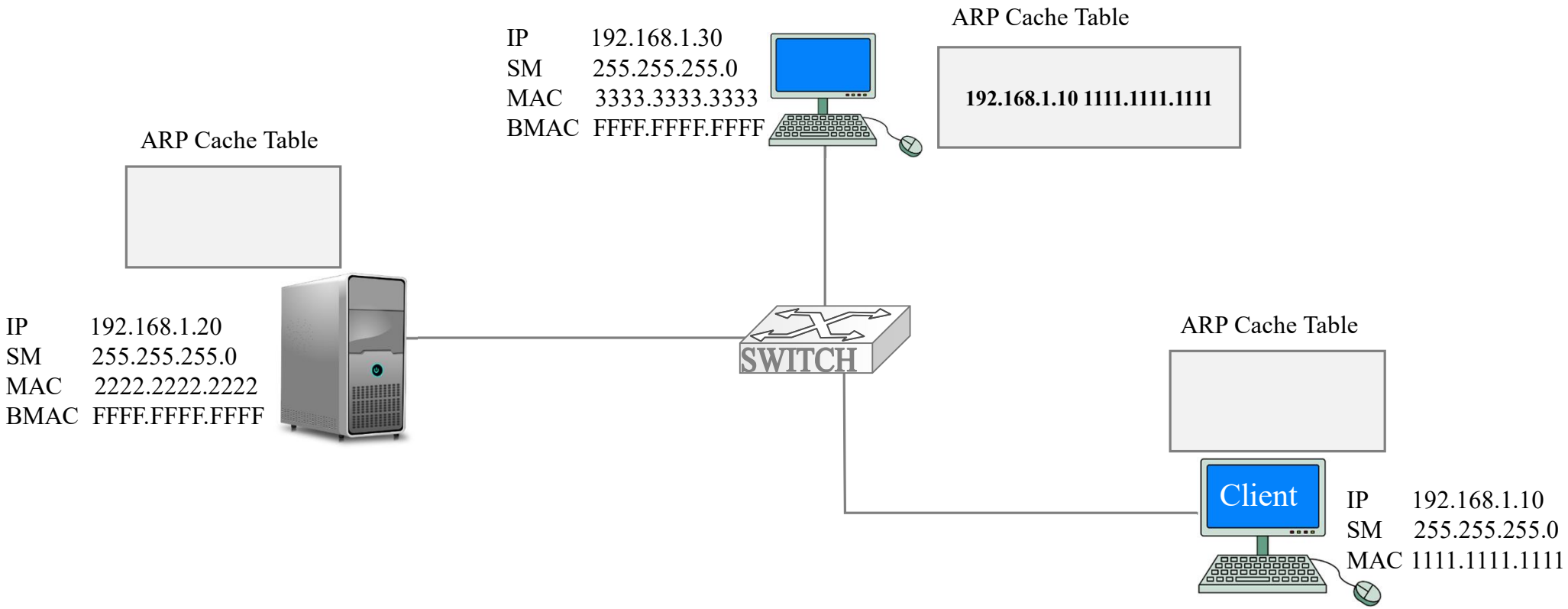
```
C:₩>arp -a

인터페이스: 192.168.35.131 --- 0x4
  인터넷 주소           물리적 주소           유형
  192.168.35.1          00-23-aa-83-11-69     동적
  192.168.35.115        38-8c-50-9a-9b-b1     동적
  192.168.35.211        04-b4-29-bf-07-09     동적
  192.168.35.255        ff-ff-ff-ff-ff-ff     정적
  224.0.0.22            01-00-5e-00-00-16     정적
  224.0.0.251           01-00-5e-00-00-fb     정적
  224.0.0.252           01-00-5e-00-00-fc     정적
  239.255.255.250       01-00-5e-7f-ff-fa     정적
  255.255.255.255       ff-ff-ff-ff-ff-ff     정적
```



ARP Request

수MAC	수MAC	Protocol	ARP Header	
FFFF.FFFF.FFFF	1111.1111.1111	0x0806	1111.1111.1111 /192.168.1.10 **0000.0000.0000/192.168.1.20**	송신지 수신지

수MAC	송MAC	송IP	수IP	송Port	수Port	전송데이타
????	1111. 1111.1111	192.168.1.10	**192.168.1.20**	50030	80	Get /www.test.com



ARP Request

수MAC	수MAC	Protocol	ARP Header
FFFF.FFFF.FFFF	1111.1111.1111	0x0806	1111.1111.1111 /192.168.1.10 송신지 **0000.0000.0000/192.168.1.20 수신지**

수MAC	송MAC	송IP	수IP	송Port	수Port	전송데이타
????	1111. 1111.1111	192.168.1.10	**192.168.1.20**	50030	80	Get /www.test.com

	ARP Header	Protocol	송MAC	수MAC
송신지 수신지	2222.2222.2222 /192.168.1.20 1111.1111.1111 /192.168.1.10	0x0806	2222.2222.2222	1111.1111.1111

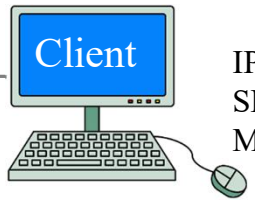
ARP Cache Table

192.168.1.10 1111.1111.1111

IP 192.168.1.20
SM 255.255.255.0
MAC 2222.2222.2222
BMAC FFFF.FFFF.FFFF

ARP Cache Table

192.168.1.20 2222.2222.2222



IP 192.168.1.10
SM 255.255.255.0
MAC1111.1111.1111

수MAC	송MAC	송IP	수IP	송Port	수Port	전송데이타
222.2222.2222	1111. 1111.1111	192.168.1.10	192.168.1.20	50030	80	Get /www.test.com